



ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

SECTION – II

Q.2	SHORT QUESTIONS	
(i)	What is meant by a Two-way analysis of Variance and an interaction?	(4)
(ii)	Three variable have in pairs simple correlation coefficients given by $r_{12} = 0.8$, $r_{13} = -0.7$ and $r_{23} = -0.9$. Find the multiple correlation coefficient $R_{1.23}$ of X_1 on X_2 and X_3 .	(4)
(iii)	(a) Define a Contingency Table. How do you determine the number of degrees of freedom in an $r \times c$ contingency table. (b) Discuss the important properties of chi-square distribution.	(4)
(iv)	What is a multiple regression? Explain the basic difference between simple regression and multiple regression.	(4)
(v)	Define and discuss the use of randomization and replication in designing an experiment.	(4)

SECTION – III

LONG QUESTIONS

Q.3	Five pennies were tossed 1000 times and the number of heads were observed as given below <table border="1"><tr><td>Number of heads</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>Frequencies</td><td>38</td><td>144</td><td>342</td><td>287</td><td>164</td><td>25</td></tr></table> Test whether a binomial distribution gives a satisfactory fit to test data by applying the chi-square goodness-of-fit-test.	Number of heads	0	1	2	3	4	5	Frequencies	38	144	342	287	164	25	(6)																												
Number of heads	0	1	2	3	4	5																																						
Frequencies	38	144	342	287	164	25																																						
Q.4	Perform the analysis of variance on the following data and analyse the treatment means using the "least significant difference" test with a 0.05 level of significance. <table border="1"><tr><th></th><th colspan="6">Treatments</th></tr><tr><th></th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th></tr><tr><td>Blocks 1</td><td>1</td><td>3</td><td>6</td><td>4</td><td>3</td><td>2</td></tr><tr><td>2</td><td>1</td><td>4</td><td>4</td><td>8</td><td>5</td><td>1</td></tr><tr><td>3</td><td>3</td><td>6</td><td>7</td><td>8</td><td>4</td><td>3</td></tr><tr><td>4</td><td>2</td><td>3</td><td>2</td><td>3</td><td>2</td><td>1</td></tr></table>		Treatments							1	2	3	4	5	6	Blocks 1	1	3	6	4	3	2	2	1	4	4	8	5	1	3	3	6	7	8	4	3	4	2	3	2	3	2	1	(6)
	Treatments																																											
	1	2	3	4	5	6																																						
Blocks 1	1	3	6	4	3	2																																						
2	1	4	4	8	5	1																																						
3	3	6	7	8	4	3																																						
4	2	3	2	3	2	1																																						

P.T.O.



Q.5	The following table shows the corresponding values of three variables X_1, X_2 and X_3. <table> <tr> <td>X_1</td><td>3</td><td>5</td><td>6</td><td>8</td><td>12</td><td>14</td></tr> <tr> <td>X_2</td><td>16</td><td>10</td><td>7</td><td>4</td><td>3</td><td>2</td></tr> <tr> <td>X_3</td><td>90</td><td>72</td><td>54</td><td>42</td><td>30</td><td>12</td></tr> </table> a) Find the regression equation X_3 on X_1 and X_2. b) Estimate X_3 when $X_1 = 10$ and $X_2 = 6$. c) Compute $R_{3.12}$ and $S_{3.12}$.	X_1	3	5	6	8	12	14	X_2	16	10	7	4	3	2	X_3	90	72	54	42	30	12	(6)
X_1	3	5	6	8	12	14																	
X_2	16	10	7	4	3	2																	
X_3	90	72	54	42	30	12																	
Q.6	Given the following data <table> <tr> <td>Y</td><td>2</td><td>5</td><td>7</td><td>8</td><td>5</td></tr> <tr> <td>X_1</td><td>8</td><td>8</td><td>6</td><td>5</td><td>3</td></tr> <tr> <td>X_2</td><td>0</td><td>1</td><td>1</td><td>3</td><td>4</td></tr> </table> (a) Calculate the estimated regression equation, (i.e $Y = a + b_1X_1 + b_2X_2$) for the above data. (b) State the meaning of the partial regression coefficients b_1 and b_2.	Y	2	5	7	8	5	X_1	8	8	6	5	3	X_2	0	1	1	3	4	(6)			
Y	2	5	7	8	5																		
X_1	8	8	6	5	3																		
X_2	0	1	1	3	4																		
Q.7	The following data obtained from a randomized complete block design with 3 treatments A, B and C and 3 blocks contain one missing observation represented by x. <table> <tr> <th>Blocks</th><th colspan="3">Treatments</th></tr> <tr> <th></th><th>A</th><th>B</th><th>C</th></tr> <tr> <td>I</td><td>5</td><td>12</td><td>15</td></tr> <tr> <td>II</td><td>7</td><td>10</td><td>14</td></tr> <tr> <td>III</td><td>8</td><td>16</td><td>x</td></tr> </table> Estimate the missing observation and prepare a table for analysis of variance.	Blocks	Treatments				A	B	C	I	5	12	15	II	7	10	14	III	8	16	x	(6)	
Blocks	Treatments																						
	A	B	C																				
I	5	12	15																				
II	7	10	14																				
III	8	16	x																				

square block RCBD

20.3 TWO-WAY ANALYSIS OF VARIANCE

When each observation is classified according to two criteria (or variables) of classification simultaneously, we use the two-way analysis of variance technique. The classified data are recorded in a table, in which the columns represent one criterion (or variable) of the classification and the rows represent the other criterion. If there are c -columns and r -rows in the table, then there will be altogether rc cells. Each cell may contain a single observation or several observations.

There are two basic forms of two-way analysis of variance, depending upon whether the two variables of classification are independent or whether they interact. Two variables (or criteria) of classification are said to interact when they together have an additive effect that they do not have individually. For example, suppose that we classify age and educational attainment with a view to their significant effect on the

Question No 11.10 (a)

$$r_{12} = 0.8 \quad r_{13} = -0.7 \quad r_{23} = -0.9$$

Multiple correlation coefficient $R_{1.23}$ of X_1 on X_2 & X_3

$$R_{1.23} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{23}r_{13}}{1 - r_{23}^2}} = \sqrt{\frac{(0.8)^2 + (-0.7)^2 - 2(0.8)(-0.7)(-0.9)}{1 - (-0.9)^2}}$$

$$R_{1.23} = \sqrt{0.2291 / 0.19}$$

$$R_{1.23} = 0.8013$$

17.7.1. Co-efficient of Contingency for an $r \times c$ contingency table. The chi-square statistic shows only whether the sample data do or do not conform to the hypothesis. It does not tell anything about the strength of the association, which we sometimes desire to measure. For this purpose, Karl Pearson (1857–1936) has defined a co-efficient C , known as the *Pearson's co-efficient of mean-square contingency*, by the relation

$$C = \sqrt{\frac{\chi^2}{n + \chi^2}},$$

where n indicates sample size.

This co-efficient measures the strength of the association dependence of two variables of classification of a contingency table. When there is complete independence, $C=0$. When the classifications are perfectly associated, $C = \sqrt{(k-1)/k}$, where k is smaller of r and c . Thus C lies between 0 and $\sqrt{(k-1)/k}$. In a contingency table, the maximum value of C is $\sqrt{(2-1)/2} = 0.707$.

11.1 INTRODUCTION

The technique of simple regression which involves one dependent variable and one independent variable is often inadequate in most real-world situations where a variable depends upon two or more independent variables or regressors. For example, the yield of a crop depends upon the fertility of the land, fertilizer applied, rainfall, quality of seed, etc. Likewise, the systolic blood pressure of a person depends upon one's weight, age, etc. In such cases, the technique of simple regression may be expanded to include several independent variables. (A regression which involves two or more independent variables is called a *multiple regression*. Thus, in case of multiple linear

23.2.1. Randomization. (The first principle of an experimental design is *randomization*, which is a random process of assigning treatments to the experimental units.) The random process implies that every possible allotment of treatments has the same probability. An *experimental unit* is the smallest division of the experimental material and a *treatment* means an experimental condition whose effect is to be measured and compared. The purpose of randomization is to remove bias and other sources of extraneous variation, which are not controllable. Another advantage of randomization (accompanied by replication) is that it forms the basis of any valid statistical test. Hence the treatments must be assigned *at random* to the experimental units. Randomization is usually done by drawing numbered cards from a well-shuffled pack of cards, or by drawing numbered balls from a well-shaken container or by using tables of random numbers.

23.2.2. Replication. The second principle of an experimental design is *replication*; which is a repetition of the basic experiment. In other words, it is a complete run for all the treatments to be tested in the experiment. In all experiments, some variation is introduced because of the fact that the experimental units such as individuals or plots of land in agricultural experiments, cannot be physically identical. This type of variation can be removed by using a number of experimental units. We therefore perform the experiment more than once, i.e., we repeat the basic experiment. An individual repetition is called a *replicate*. The number, the shape and the size of replicates depend upon the nature of the experimental material. A replication is used

- (i) to secure more accurate estimate of the experimental error, term which represents the differences that would be observed if the same treatments were applied several times to the same experimental units;

- (ii) to decrease the experimental error and thereby to increase *precision*, which is a measure of the variability of the experimental error; and
- (iii) to obtain more precise estimate of the mean effect of a treatment, since $\sigma_{\bar{y}}^2 = \frac{\sigma^2}{n}$, where n denotes the number of replications.

Example 17.10. Five pennies were tossed 1,000 times and the number of heads were observed as given below:

Number of heads	0	1	2	3	4	5
Frequencies	38	144	342	287	164	25

Test whether a binomial distribution gives a satisfactory fit to these data. (P.U., B.A./B.Sc. 1960)

*always test
two
ided*

(i) We state our hypotheses as

H_0 : The population distribution is a binomial with $n=5$, but with parameter p unspecified, and

H_1 : The population distribution is not a binomial with $n=5$.

(ii) We choose the significance level at $\alpha = 0.05$.

(iii) We use the test-statistic

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$$\chi^2 = \sum \frac{(o_i - e_i)^2}{e_i}$$

which, if H_0 is true, has an approximate χ^2 -distribution with degrees of freedom = $k-1$ -number of estimated parameters. A binomial distribution has two parameters n and p but n is specified ($n=5$). We have to estimate the value of one parameter

p from the sample data. Therefore the degrees of freedom = $6 - 1 - 1 = 4$ ($\because k = 6$ categories).

we only estimated p , and n is given.

(iv) The critical region is $\chi^2 \geq \chi_{0.05, (4)}^2 = 9.49$.

(v) **Computations.** To estimate the value of p , we first compute the mean number of heads, $\bar{x}$. Thus

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{2470}{1000} = 2.47.$$

$$2353 \quad | \quad 15 \quad | \quad 100$$

Theoretically, $\bar{x} = np$, so that $\hat{p} = \frac{\bar{x}}{n} = \frac{2.47}{5} = 0.494$.

Hence the expected (fitted) frequencies are the terms in the binomial expansion of $1,000(0.506 + 0.494)^5$, which are given in the column headed e_i . Next, we calculate the value of χ^2 as follows:

Number of heads	Observed f_i (o_i)	Expected f_i (e_i)	$o_i - e_i$	$(o_i - e_i)^2$	$(o_i - e_i)^2 / e_i$
0	38	33.2	4.8	23.04	0.69
1	144	161.9	-17.9	320.41	1.98
2	342	316.2	25.8	665.64	2.15
3	287	308.7	-21.7	470.89	1.53
4	164	150.7	13.3	176.89	1.17
5	25	29.4	-4.4	19.36	0.66
Total	1,000	1,000.1	---	---	$\chi^2 = 8.18$

(vi) **Conclusion.** Since the calculated value of $\chi^2 = 8.18$ does not fall in the critical region, we therefore are unable to reject our null hypothesis. We may accept the hypothesis that the distribution of the number of heads is a binomial distribution and conclude that the fit of data is good.

Even

Example 20.7. Perform the analysis of variance on the following data and analyse the treatment means using the *LSD* test with a 0.05 level of significance.

2019

	Treatments					
	1	2	3	4	5	6
1	1	3	6	4	3	2
2	1	4	4	8	5	1
3	3	6	7	8	4	3
4	2	3	2	3	2	1

- (i) We formulate our null and alternative hypotheses as

$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5 = \mu_6$, i.e. the six treatment means are equal, and

H_1 : Not all six treatment means are equal.

- (ii) The significance level is set at $\alpha = 0.05$.

- (iii) We use the F -test to accept or reject H_0 . If H_0 is rejected, the LSD-test is used to analyse the treatment means.

- (iv) The computations are presented in the following ANOVA-Table.

Source of Variation	d.f.	Sum of Squares	Mean Square	F
Between Treatments	5	51.83	10.37	$F_1 = \frac{10.37}{1.57} = 6.61$
Between Blocks	3	28.50	9.50	..
Error	15	23.50	$s^2 = 1.57$	
Total	23	103.83	--	--

- (v) The critical region for treatments is $F_1 \geq F_{0.05}(5, 15) = 2.90$.

- (vi) **Conclusion.** We reject the null hypothesis of equal treatment means as the computed value of $F_1 = 6.61$ exceeds the table value of $F = 2.90$. As H_0 is rejected, using F -test, we therefore apply the LSD test to find out which means differ from each other.

The Least-Significant Difference is given by the relation

$$LSD = t_{0.025, (15)} \sqrt{\frac{2s^2}{r}}; \text{ where } s^2 = 1.57 \text{ and } r = 4.$$

$$= 2.13 \sqrt{\frac{2 \times 1.57}{4}} = 1.89$$

Arranging the treatment means ($\bar{x}_i$) in ascending order of magnitude, and drawing a line under pair of adjacent means (or sets of means) that are not significantly different, we have

$\bar{x}_1$	$\bar{x}_6$	$\bar{x}_5$	$\bar{x}_2$	$\bar{x}_3$	$\bar{x}_4$
1.75	1.75	3.50	4.00	4.75	5.75

The significantly different pairs are immediately observed from this presentation.

$$\sum X_1 X_3 = b_{3.12} \sum X_1 + b_{31.2} \sum X_1^2 + b_{32.1} \sum X_1 X_2,$$

$$\sum X_2 X_3 = b_{3.12} \sum X_2 + b_{31.2} \sum X_1 X_2 + b_{32.1} \sum X_2^2.$$

The calculations needed to find b 's are shown in the table below:

X_1	X_2	X_3	X_1^2	X_2^2	X_3^2	$X_1 X_2$	$X_2 X_3$	$X_1 X_3$
3	16	90	9	256	8100	48	1440	270
5	10	72	25	100	5184	50	720	360
6	7	54	36	49	2916	42	378	324
8	4	42	64	16	1764	32	168	336
12	3	30	144	9	900	36	90	360
14	2	12	196	4	144	28	24	168
48	42	300	474	434	19008	236	2820	1818

Substituting the sums in the normal equations, we get

$$6b_{3.12} + 48b_{31.2} + 42b_{32.1} = 300$$

$$48b_{3.12} + 474b_{31.2} + 236b_{32.1} = 1818$$

$$42b_{3.12} + 236b_{31.2} + 434b_{32.1} = 2820$$

Solving them simultaneously, we obtain

$$b_{3.12} = 61.3993, b_{31.2} = -3.6461 \text{ and } b_{32.1} = 2.5385$$

Hence the desired estimated regression equation of X_3 on X_1 and X_2 is

$$\hat{X}_3 = 61.40 - 3.65X_1 + 2.54X_2.$$

(b) When $X_1 = 10$ and $X_2 = 6$, we get

$$X_3 = 61.40 - 3.65(10) + 2.54(6)$$

$$= 61.40 - 36.5 + 15.24 = 40.$$

(c) The multiple correlation co-efficient is the positive square root of $R^2_{3.12}$, where

$$\begin{aligned} R^2_{3.12} &= \frac{\sum(\hat{X}_3 - \bar{X}_3)^2}{\sum(X_3 - \bar{X}_3)^2} \\ &= \frac{b_{3.12} \sum X_3 + b_{31.2} \sum X_1 X_3 + b_{32.1} \sum X_2 X_3 - (\sum X_3)^2/n}{\sum X_3^2 - (\sum X_3)^2/n} \end{aligned}$$

Hence $R_{3,12} = \sqrt{0.9855} = 0.99$.

Alternatively:

$$\text{Now, } \bar{X}_1 = \frac{\sum X_1}{n} = \frac{48}{6} = 8; \bar{X}_2 = \frac{\sum X_2}{n} = \frac{42}{6} = 7;$$

$$\bar{X}_3 = \frac{\sum X_3}{n} = \frac{300}{6} = 50.$$

$$S_1 = \sqrt{\frac{\sum X_1^2}{n} - \left(\frac{\sum X_1}{n}\right)^2} = \sqrt{\frac{474}{6} - \left(\frac{48}{6}\right)^2}$$

$$= \sqrt{79 - 64} = \sqrt{15} = 3.87;$$

$$S_2 = \sqrt{\frac{\sum X_2^2}{n} - \left(\frac{\sum X_2}{n}\right)^2} = \sqrt{\frac{434}{6} - \left(\frac{42}{6}\right)^2}$$

$$= \sqrt{72.33 - 49} = \sqrt{23.33} = 4.83;$$

$$S_3 = \sqrt{\frac{\sum X_3^2}{n} - \left(\frac{\sum X_3}{n}\right)^2} = \sqrt{\frac{19008}{6} - \left(\frac{300}{6}\right)^2}$$

$$= \sqrt{3168 - 2500} = \sqrt{668} = 25.84;$$

and $r_{12} = \frac{\sum X_1 X_2 - n \bar{X}_1 \bar{X}_2}{\sqrt{[\sum X_1^2 - n \bar{X}_1^2] [\sum X_2^2 - n \bar{X}_2^2]}}$

$$= \frac{236 - 6(8)(7)}{\sqrt{[474 - 6(64)] [434 - 6(49)]}} = \frac{236 - 336}{\sqrt{(90)(140)}}$$

$$= \frac{-100}{112.25} = -0.89;$$

$$r_{23} = \frac{\sum X_2 X_3 - n \bar{X}_2 \bar{X}_3}{\sqrt{[\sum X_2^2 - n \bar{X}_2^2] [\sum X_3^2 - n \bar{X}_3^2]}}$$

$$= \frac{2820 - 6(7)(50)}{\sqrt{[434 - 6(49)] [19008 - 6(2500)]}} = \frac{2820 - 2100}{\sqrt{(140)(4008)}}$$

$$= \frac{720}{749.08} = 0.96;$$

$$r_{31} = \frac{\sum X_1 X_3 - n \bar{X}_1 \bar{X}_3}{\sqrt{[\sum X_1^2 - n \bar{X}_1^2] [\sum X_3^2 - n \bar{X}_3^2]}}$$

$$= \frac{1818 - 2400}{\sqrt{(90)(4008)}} = \frac{-582}{600.6} = -0.97.$$

(a) The linear regression equation of x_3 on x_1 and x_2 is

$$\frac{x_3}{S_3} = \left(\frac{r_{32} - r_{13}r_{12}}{1 - r_{12}^2} \right) \frac{x_2}{S_2} + \left(\frac{r_{31} - r_{23}r_{12}}{1 - r_{12}^2} \right) \frac{x_1}{S_1},$$

where $x_1 = X_1 - \bar{X}_1$, $x_2 = X_2 - \bar{X}_2$ and $x_3 = X_3 - \bar{X}_3$.

Substituting the values, we get

$$\frac{x_3}{25.84} = \left(\frac{(0.96) - (-0.97)(-0.89)}{1 - (-0.89)^2} \right) \frac{x_2}{4.83} + \left(\frac{-0.97 - (0.96)(-0.89)}{1 - (-0.89)^2} \right) \frac{x_1}{3.87}$$

Or
$$\frac{x_3}{25.84} = \left(\frac{0.0967}{0.2079} \right) \frac{x_2}{4.83} + \left(\frac{-0.1156}{0.2079} \right) \frac{x_1}{3.87}$$

Or
$$x_3 = 2.49x_2 - 3.71x_1$$

Or
$$(X_3 - 50) = 2.49(X_2 - 7) - 3.71(X_1 - 8)$$

Or
$$X_3 = 62.25 - 3.71X_1 + 2.49X_2.$$

(b) When $X_1 = 10$, and $X_2 = 6$, we get

$$X_3 = 62.25 - 3.71(10) + 2.49(6) = 40$$

11.3. (a) The estimated multiple linear regression equation is

$$\hat{Y} = a + b_1X_1 + b_2X_2,$$

where a , b_1 and b_2 are the least-squares estimates of the parameters α , β_1 and β_2 .

The three normal equation are

$$\sum Y = na + b_1\sum X_1 + b_2\sum X_2$$

$$\sum X_1Y = a\sum X_1 + b_1\sum X_1^2 + b_2\sum X_1X_2$$

$$\sum X_2Y = a\sum X_2 + b_1\sum X_1X_2 + b_2\sum X_2^2$$

The calculations needed to find a , b_1 and b_2 are shown below:

Y	X_1	X_2	X_1^2	X_2^2	X_1Y	X_2Y	X_1X_2
2	8	0	64	0	16	0	0
5	8	1	64	1	40	5	8
7	6	1	36	1	42	7	6
8	5	3	25	9	40	24	15
5	3	4	9	16	15	20	12
27	30	9	198	27	153	56	41

Substituting the sums in the normal equations, we get

$$5a + 30b_1 + 9b_2 = 27$$

$$30a + 198b_1 + 41b_2 = 153$$

$$9a + 41b_1 + 27b_2 = 56$$

Solving them simultaneously, we obtain

$$a = 4.49, b_1 = -0.04 \text{ and } b_2 = 0.64$$

Hence the desired estimated multiple linear regression is

$$\hat{Y} = 4.49 - 0.04X_1 + 0.64X_2$$

Example 23.5. Estimate the missing value (y_0) from the data given below by applying the covariance technique:

(2019)

Blocks	Treatments		
	A	B	C
I	5	12	15
II	7	10	14
III	8	16	y_0

In order to carry out the analysis of covariance, we insert zero for the missing observation and pair the observations with the covariate composed of zeros and a one as shown in the table.

Blocks		Treatments						B_i	B_i'	$B_i'^2$
		A		B		C				
		X	Y	X	Y	X	Y			
I	0	5	0	12	0	15	0	32	1024	
II	0	7	0	10	0	14	0	31	961	
III	0	8	0	16	1	0	1	24	576	
T_j, T_j'	0	20	0	38	1	29	1	87	2561	
$T_j'^2$	400		1444		841		2685			

The sum of squares and products are computed below:

Source of Variation	Σx^2	Σy^2	Σxy
C.F.	$\frac{(1)^2}{9} = \frac{1}{9}$	$\frac{(87)^2}{9} = 841$	$\frac{1 \times 87}{9} = 9.67$
Total	$1 - \frac{1}{9} = \frac{8}{9}$	$1059 - 841 = 218$	$0 - 9.67 = -9.67$
Treatment	$\frac{1}{3} - \frac{1}{9} = \frac{2}{9}$	$\frac{2685}{3} - 841 = 54$	$\frac{1 \times 29}{3} - 9.67 = 0$
Block	$\frac{1}{3} - \frac{1}{9} = \frac{2}{9}$	$\frac{2561}{3} - 841 = 13$	$\frac{24}{3} - 9.67 = -1.67$
Error	(By diff.) $= \frac{4}{9}$	$= 151$	$= -8.00$

The analysis of covariance table is then set up as below:

Source of Variation	d.f.	Sum of Squares and Products			Regression Co-efficient
		$\sum x^2$	$\sum y^2$	$\sum xy$	
Block	2	2/9	13	-1.67	$b = \frac{-8.00}{4/9} = -18$
Treatment	2	2/9	54	0	
Error	4	4/9	151	-8.00	
Total	8	8/9	218	-9.67	

Hence the estimate of the missing observation = $-b = 18$.